

the efficacy of a laser weapon.

The laser beam has to propagate over long atmospheric paths and the resulting propagation effects control the overall effective range or lethality range of the weapon system. Propagation effects mainly include diffraction, absorption and turbulence.

Diffraction effect tends to spread out the laser energy as the laser beam propagates, and the amount of spreading is proportional to wavelength to beam aperture diameter ratio. This limits the smallest spot diameter to which the laser beam can be focused.

Scattering is due to atmospheric constituents and results in loss of laser energy from the beam path in other directions. Longer wavelengths have relatively lower scattering losses.

Absorption in atmospheric constituents also causes loss of energy, consequently decreasing the laser power reaching the target end. There is very strong dependence of absorption parameter on location, time of year and weather pattern.

Turbulence causes the beam to spread, resulting in an increase in laser spot size and reduction in power density at the target end. Turbulence is a strong function of altitude, wind speed and other atmospheric parameters.

Thermal blooming is another parameter that causes laser beam aberration. This arises out of heating effects produced by absorption, leading to refractive index gradient in the laser beam path. This further leads to the formation of negative lens and causes the laser beam to spread. Thermal blooming effects can be ignored for rapidly-moving targets such as airborne targets.

HPL sources

An HPL source is an important constituent of a laser based DEW system. Different types of laser sources have been exploited and

experimented with for DEW application. Gas dynamic lasers and chemical lasers, HF/DF lasers in particular, were used to generate the high power levels required for a DEW application in 1970s and 1980s, as the technology of these lasers was well-established.

Due to advances in solid-state laser and fibre-laser technologies, these are considered today as the most potent laser source candidates for deployable laser weapon systems. Chemical oxy-iodine laser is also very much in contention due to its shorter wavelength of operation, particularly for generating power in megawatts.

Critical requirements

Laser power and beam quality are the two most important laser source parameters that decide the high-energy laser weapon's lethal capability. Beam quality determines the smallest achievable focused laser spot size. This together with laser power available at the target site determines power density and, hence, lethality.

It is evident that for maximizing power density at the target, one would prefer a shorter wavelength, high beam quality, larger beam diameter aperture size and low atmospheric transmission losses.

Shorter wavelength is preferred as it not only produces a higher power density, target coupling efficiency is also higher at shorter wavelengths. For the same laser power and transmitting telescope aperture, one-micron HPL (chemical oxy-iodine laser, solid-state lasers) would have an operational range that is approximately ten times that of a ten-micron HPL (CO_2 laser) for a given power density requirement at the target.

Beam quality is essentially a measure of how tightly the laser beam can be focused to form a small and intense spot of light on a distant target. A beam quality of

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